Auto Co-relation Test for Random Nubers

- Consider a sequence of 30 numbers. Test weather 3rd 8th 13th ... Numbers in sequence are auto co-related with **α = 0.05 and $Z_{0.025}$ =1.96**

0.74	**0.6**	**0.87**	**0.64**	**0.07**	**0.22**
0.35	**0.24**	**0.49**	**0.39**	**0.41**	**0.37**
0.77	**0.77**	**0.74**	**0.37**	**0.89**	**0.47**
0.51	**0.11**	**0.37**	**0.95**	**0.49**	**0.04**
0.86	**0.25**	**0.42**	**0.19**	**0.41**	**0.25**

Find M

- Find M
- $i+(M+1)m = N$
- $i=3$
- $m=5$ (Gap between random number)
- $N=30$ (Total Random number)

$(M+1)=(N-i)/m$

$M=\{(N-i)/m\} - 1$

M= 4.4 >> 4

Find Rho (**ρ**)

$$\boldsymbol{\rho} = \frac{1}{M+1}\left[\sum_{k=0}^{M} R_{i+km}\, R_{i+(k+1)m}\right] - 0.25$$

Find Sigma (**σ**)

- $\boldsymbol{\sigma} = \frac{\sqrt{13M+7}}{12(M+1)}$

Find Z_0

- $Z_0 = \frac{\boldsymbol{\rho}}{\boldsymbol{\sigma}}$

Checking Z_0 Z_α

- If $-Z_{\alpha/2} <= Z_0 <= Z_{\alpha/2}$
- Hypothesis is accepted and numbers are auto corelated.